

Motivation

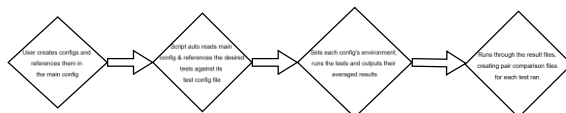
- Started as an introductory script made to calculate averages on three specific OSU-micro-benchmark tests. Was created to get acquainted with the environment and processes used by NOWLAB during onboarding
- Expanded functionality to assist benchmark testing and reduce manual calculation workload for other lab members.

```
# OSU MPI Put Latency Test v7.0
# Window creation: MPI_Win_allocate
# Synchronization: MPI_Win_Flush
# Datatype: MPI_CHAR
# Size      Latency (us)
1          0.24
2          0.24
4          0.24
8          0.24
16         0.24
32         0.24
64         0.24
128        0.24
256        0.24
512        0.24
1024       0.24
2048       0.25
4096       0.26
8192       0.30
16384      0.52
32768      0.85
65536      1.43
131072     2.79
262144     5.92
524288     26.43
1048576    53.10
2097152    100.69
4194304    205.95
```

Fig 1. Standard output for test

Strategy & Technologies

- Iterative implementation**
Allows for quicker update turnaround – by adding functionality in an iterative manner, I was able to make more frequent expansions to the functionality, and reduced time doing major code refactoring
- Config files**
Users can modify the desired test inputs without having to edit the original script or manually pass in long strings of parameters and set configurations each run. Localizes changes for batch testing to a simple .yaml file
- Just bash**
The script uses snippets of awk but was written in bash to be as simple and portable as possible – default availability on Linux systems removes the need for any extra imports to get the script to run.
- Job scheduling via Slurm**
Otherwise known as the "Simple Linux Utility for Resource Management" is SLURM is an open-source job scheduling system. It allocates resources and provides structure and management tools for clusters of compute nodes



Capabilities

Multi-system support:

Config files have ability to specify command execution style, allowing support for various launchers like ibrun, srun, mpirun...

Modularization:

The used modules are integrated into the script, allowing users to run without stopping to configure the environment in-between.

Ex. Comparisons of CPU and GPU require different lmod modules to be loaded into the execution space.

```
config:
  BASE_DIR: "home/user_dir"
  RESULTS_DIR: "/home/user_dir/results_dir"
  AVG_RUNS: 5
  sources:
    - ../config/mvapich_cpu.yaml
    - ../config/mvapich_cuda.yaml
  benchmarks:
    - osu_latency
    - osu_bw
    - osu_bibw
    - osu_allgather
```

Fig 2. Head Config

Environment Variables:

A central feature of the script is the ability to benchmark systems across different CVAR* sets – each configuration might only have one or two differences and being able to quickly find the change in performance between configurations eliminates even more manual calculations!
*environment variables

```
install:
  name: "default-mvapich"
  omb_source: "/path/to/osu-micro-benchmarks"
  omb_version: "ver 7.5"
  launcher_path: "/path/to/desired/mpiexec"
  launcher_default: null
  task_type: "CPU"
  gpu_config: null
  modules: null
  cvars_file: null
  cvars_list:
    - FI_PROVIDER="mverbs;ofi-ucx"
```

Fig 3. Individual Installation Config

Results:

MSG size (MPI Char)	CPU-run (avg MB/s)	GPU-run (avg MB/s)	Percent difference	Increased Bandwidth GPU comp. CPU
32	43.85	59.224	.2983%	1.35x increase
64	77.87	114.77	.3830%	1.47x increase
128	169.164	221.776	.2691%	1.31x increase
256	323.556	374.626	.1462%	1.16x increase
16824	12559.476	9080.958	.3214%	0.72x decrease
4194304	24477.75	22197.288	.0977%	0.91x decrease

Tab 1. Selections from bandwidth Test – CPU vs. GPU c onfigs, examining the GPU bottleneck

Quick Guide

High Performance Computing

The use of super computers and clusters of compute nodes to efficiently calculate solutions to problems that a standard computer cannot. Typically utilizes parallelization to execute several calculations at the same time!

MVAPICH

MVAPICH is a high performance implementation of the Message Passing Interface supporting transfers over multiple high-performance networks.

What is MPI?

The Message Passing Interface is the standard that defines how implementations like MVAPICH structure the communication and information transfer between computational nodes (one sided communications, point-to-point etc.) Their flexibility provides a solid base to build from; allowing portability and optimization on new systems!

OSU Micro Benchmarks (OMB)

A standardized performance measurement test suite for MPI operations

NowLab

The Network Based Computing Lab at OSU – developers of MVAPICH and OMB

Future Work

- Separation of analysis step
Allow users to run only the test or analysis section – can allow for easier cross system data comparison and elimination of need for hand calculation
- More configuration testing
Add to list of currently tested compatible system hardware
- Job scheduling variance
Create modules for use with non SLURM schedulers

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